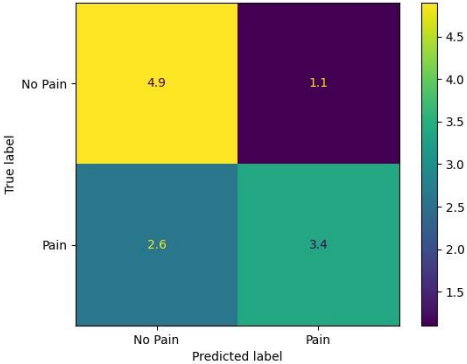
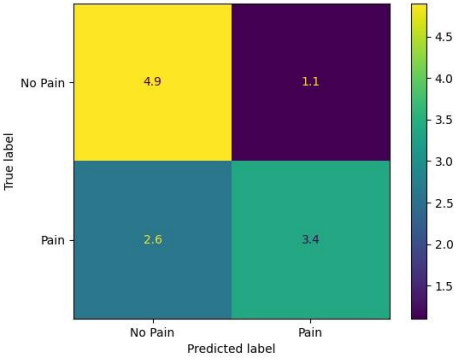
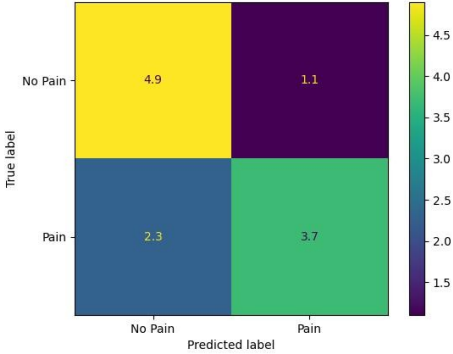


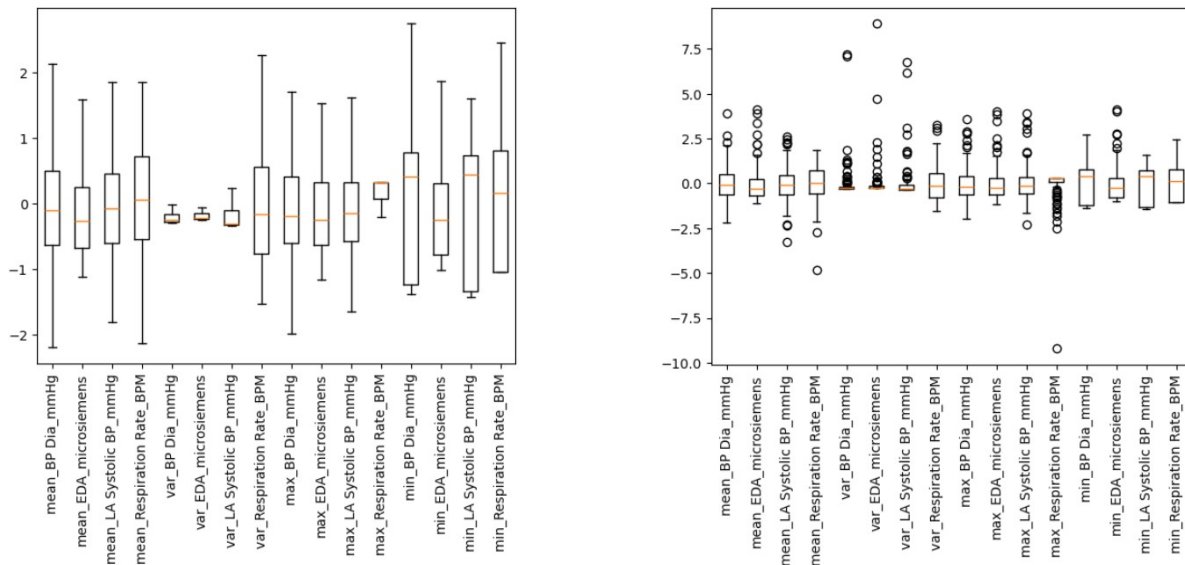
Project 2: Pain Recognition

- 1. I went with a logistic regression for this task. Initially, I chose it just as a quick baseline because this is a binary classification, but I found the results were very robust and comparable to non-linear models like random forest. For some data types, I even found the regressor performed better. Observing this performance, I decided to keep it for simplicity, speed, and theoretical interpretability regarding each of the features.
- 2. My model was most accurate when using all the data (72%), but if this is asking for a single feature then the blood pressure measures (diastolic and systolic) were the best at 69% each. Blood pressure has a known correlation with pain – it is a symptom of the “fight or flight” system, improving the body’s ability to respond to stress or physical demand. This supports the high accuracy of these metrics.

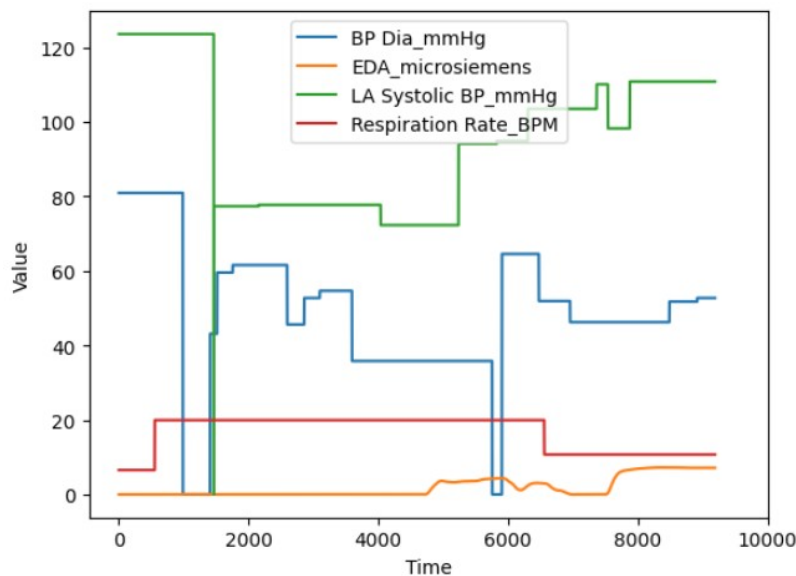
	Accuracy	Precision	Recall	Confusion Matrix
Diastolic	69.2%	79.6%	56.7%	
Systolic	69.2%	79.8%	56.7%	

All Data	71.7%	77.5%	61.7%	
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3. My fusion features performed better than any individual feature. This likely works because each data type has different strengths and weaknesses; considering all of them makes up for each one's blind spots and smooths out their errors. This supports what we've been learning in class about the importance of multimodal data. Human systems are multifaceted, and taking in more views of the same event (i.e., context) leads to better predictions.
4. I scaled each feature before making boxplots, as they all have different units. I found that the features were generally not highly variable, having their interquartile range contained within one standard deviation of their mean and their medians near the mean. Upper and lower quartiles did not extend much beyond two standard deviations for even the metrics with the widest spread. Some features had very little variation, namely the variance of blood pressure metrics and of EDA. That being said, nearly all the features had several outliers, commonly approaching +/- five deviations and in some cases more than eight deviations. For most features, these are upper outliers, indicating a right skew.



5. I plotted the metrics of a single subject's pain response, and saw that the most highly varying signals were diastolic and systolic blood pressure. This aligns with the results showing that blood pressure is a strong indicator of pain as discussed above. Respiration and EDA varied little in comparison, exhibiting mostly flat lines. This backs up that blood pressure is closely related to pain, and experiencing pain is likely to cause fluctuations in blood pressure.



6. My intuition would be that pain is associated with facial expressions, as one may be inclined to grimace or similar when experiencing acute pain. That being said, facial

reactions vary widely among individuals, so it may be difficult to actually train a pain recognition model based on this. Datasets have been constructed involving 2D and 3D facial data in conjunction with physiological data, and these have achieved emotion recognition accuracy of up to 75% [1], so it is not impossible that facial data could be used for pain as well.

- [1] Z. Zhang et al., "Multimodal spontaneous emotion corpus for human behavior analysis," IEEE Xplore, Jun. 01, 2016. <https://ieeexplore.ieee.org/document/7780743>